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acousticDE: A diffusion equation model package for room acoustics simulations

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Summary

Studying the acoustic properties of enclosed spaces is important for improving the quality of environments and reducing noise induced health effects. With the rise of numerical room acoustics modelling, tools are needed to predict sound behaviour in enclosed spaces. acousticDE is an open-source software for simulating room acoustics using the diffusion equation approach. The software is designed to be easy to use and suitable for research applications, consultancy work and educational purposes. In practice, it can be used for evaluating design choices during early architectural design stages, optimising acoustic treatment layouts or testing performance of rooms intended for speech, music or general occupancy.

The method implemented in acousticDE - the diffusion equation method - offers a balance between computational efficiency and flexibility with regards to the type of spaces to be addressed. To simulate the sound field in a room, acousticDE requires a digital 3D model of the geometry, as well as material properties describing the acoustic boundaries. Once configured, acousticDE can incorporate sound sources, compute the distribution of sound energy throughout the space, and provide key room acoustics parameters relevant to practical acoustic assessments.

Statement of need

Room acoustics simulation is essential for researchers and for practitioners to accurately predict and design the acoustics inside a room. However, existing tools often face a trade off between (1) accuracy and efficiency in the calculation method, and (2) transparency in the software functionalities and operabilities. Traditional statistical methods, such as the Sabine and Eyring calculations, provide a fast global estimate of the acoustic parameters, but these are based on unrealistic assumptions that are only valid for certain rooms and for certain frequency range (Kuttruff, 2019). More accurate approaches are available including both commercial and open-source software. However, commercial tools often lack transparency, as it is difficult to understand how the calculations are performed internally. Available tools are mostly based on common approaches for running high frequency simulations, i.e. geometrical acoustics methods. Additionally, full wave-based approaches can achieve higher accuracy but require significant computational resources, making them impractical for larger spaces.

acousticDE was developed to address both these gaps. It is an open-source software package designed for the simulation of room acoustics using the diffusion equation model (DE), targeting researchers, practitioners and educators seeking a transparent and efficient method to high-frequency room acoustics.

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State of the field

The diffusion equation model has emerged as an effective compromise for room acoustics simulations in the high frequency range, providing physically meaningful spatial and temporal distributions of acoustic energy while remaining computationally efficient (Foy et al., 2016; Mou et al., 2023; Sü Gül et al., 2020). However, despite its demonstrated usefulness in the literature, accessible and user-friendly open-source software implementations of the acoustic diffusion equation model do not exist (Hornikx et al., 2024) or are somewhat limited. In fact, most implementations of the diffusion equation are embedded in commercial tools (e.g. COMSOL or Femlab toolbox for Matlab), where room acoustics simulation are not the primary focus. An additional open source implementation exists, but it is currently at its initial experimental stage, lacks documentation and is written in GNU programming language (Environmental Acoustics Research Unit, 2016).

Compared to the available software, `acousticDE` distinguishes itself by providing two different numerical methods for implementing the diffusion equation within a Python programming environment: the Du Fort&Frankel Finite Difference Method (FDM) (Navarro et al., 2012) and the Finite Volume Method (FVM) (Pagán Muñoz, 2019). The FDM is suitable for parallelepiped shapes and the room is discretised into a 3D grid of points where the energy density is calculated. The FVM supports arbitrary 3D geometry, through a volumetric tetrahedral mesh. The dual approach allows the user to select the appropriate balance between flexibility and efficiency, depending on the room type and the dimension of the discretization. Compared to other tools for the high frequency range, such as pyroomacoustics (Scheibler et al., 2018) and sound particle tracing (Picaut & Fortin, 2012), the DE model is easy to use and understand and has a low computational time, allowing users to quickly simulate room acoustic problems. However, for non-proportional geometries or highly absorptive spaces, the diffusion equation does not reach the level of accuracy of more detailed (but computationally more costly) acoustics methods (geometrical acoustics or wave-based method). It is instead accurate for compact spaces with high reflective surfaces.

In addition to standard room-acoustics parameters, `acousticDE` can generate impulse responses and support auralization, enabling perceptual audio demonstrations of simulated acoustic conditions. Comprehensive and well-structured documentation accompanies the software.

Method

The diffusion equation method, originally introduced by Ollendorff (Ollendorff, 1969) and later refined (Picaut et al., 1997), overcomes some of the limitations of statistical room acoustics, which is grounded in the diffuse field assumption (Sabine, 1922). Compared to wave-based approaches (Hamilton & Bilbao, 2017; Wang et al., 2019), ray- and beam-tracing algorithms (Kuttruff, 2019), or the image source method (Kuttruff, 2019), the DE method provides a computationally lightweight software for room acoustics simulations, research and practice. The method is generally more accurate than Sabine- or Eyring-based estimates while remaining computationally inexpensive.

The diffusion equation method estimates how acoustic energy is distributed over time and space within a specific room. The modelling method is based on solving the partial differential diffusion equation, which is applicable in the high-frequency range (above the Schroeder frequency) and it assumes only diffusely reflecting boundaries following Lambert's law (Valeau et al., 2006). The diffusion equation method is based on the energetical approach, where sound has undergone multiple reflections and scattering in an environment. It represents the late stage of wave propagation, where the energy, after being reflected multiple times, is spread out across the space, following the diffusion process of gas and heat in a medium. Surface absorption, atmospheric absorption, and diffusion are included. Limitations include the omission of diffraction, early reflections, partially scattering surfaces, and reduced accuracy at

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high average absorption coefficients.

The governing diffusion equation and boundary condition are:

$$\frac{\partial w(\mathbf{r},t)}{\partial t} = -D\boldsymbol{\nabla}^2 w(\mathbf{r},t) - mcw(\mathbf{r},t) + q(\mathbf{r},t)$$

$$-D\frac{\partial w}{\partial n} = hw$$

where $w(\mathbf{r},t)$ is the energy density at each position $\mathbf{r}$ and at time t, m is the atmospheric attenuation coefficient, c is the speed of sound, $q(\mathbf{r},t)$ is the source term and h is the boundary absorption term. Several methodologies propose expressions for h based on different assumptions on how to treat the surface absorption coefficients of the room (Jing & Xiang, 2007, 2008; Picaut, 2002). The diffusion coefficient D, with units m^2/s, indicates how quickly sound diffuses around the room. This depends on the mean free path of the room.

From the energy density, the reverberant field Sound Pressure Level (SPL) can be calculated as follows:

$$\text{SPL}(\mathbf{r},t) = 10\log_{10}\left(\frac{\rho c^2 w(\mathbf{r},t)}{p_{\text{ref}}^2}\right)$$

where ρ is the density of air in [kg/m^3] and p_{ref} is equal to 2×10^{-5} Pa (Navarro & Escolano, 2015).

This method allows to calculate acoustics parameters in the room (e.g. sound pressure level and reverberation time). Example results are shown in the figure below and included in the Example gallery documentation.

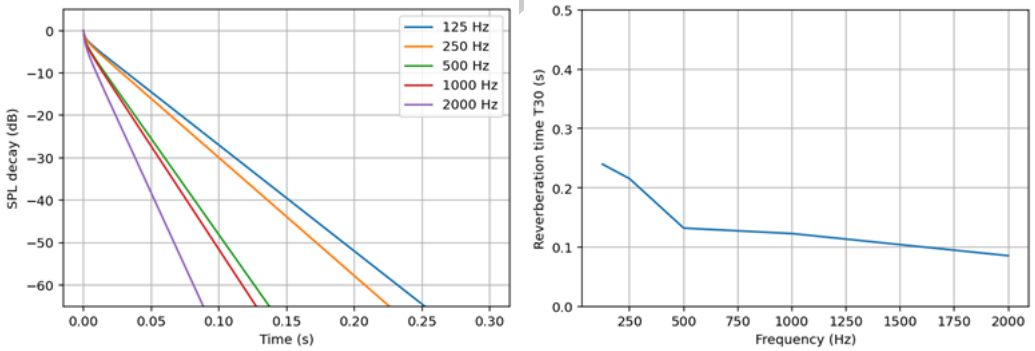


Figure 1. SPL decay (left) and reverberation time T30 (right) for a 3 x 3 x 3 [m^3] for absorption coefficients [0.3 ,0.33,0.5,0.53,0.7] at each frequency.

Software design

Design trade-offs

The design of `acousticDE` is based on architectural choices shaped by the needs of room acoustics simulation research, which are transparency, reproducibility and modularity. The software is implemented in Python, since it is open-source and widely accessible compared to licensed alternatives, i.e. Matlab. In addition, it is an easier and more understandable language for researchers and for users from different fields compared to lower level languages such as C++ and Fortran.

Method-centric modularity: Rather than implementing a single abstract solver interface, `acousticDE` is organised around self contained modules: FDM, FVM and Auralization.

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Each module exposes a clear functional API like 'run_fvm_sim(...)' and maintains its own workflow. This structure mirrors typical room-acoustics workflows and prioritises clarity over complex abstraction layers. While the software is developed as a stand-alone project, its architecture allows for potential future integration or coupling with other tools (such as the image source method for the early part of the decay), providing that a common input generation framework is used.

Workflow design: `acousticDE` adopts a functional, sequential pipeline rather than an object-oriented architecture. As an example, the FVM and Auralization pipelines are shown as follows, as these represents the most structurally demanding workflow.

1. Creation of the 3D geometry;
2. meshing of the geometry;
3. specification of inputs parameters;
4. execution of the acoustic diffusion equation simulator;
5. execution of the auralization module, based on the results of point 4.

This pipeline was intentionally chosen to make each stage of the modelling process explicit and inspectable, encouraging users to verify geometry inspect the mesh and validate the physical inputs before committing to the simulation. This also allow the user to create the geometry of the room in a different GUI based software specifically developed for 3D modelling, such as SketchUp or Blender, instead of creating the geometry by coding. The design aligns with the goal of making the modelling process explicit rather than hiding behind a 'black box'.

Functional decomposition: Simulations are executed via a single function call. Each module's main simulation engine (point 4 of the pipeline before mentioned) is organised in a set of modular, sequential function calls. Each function has its own task and returns data that are needed by the next function in the next stage of the simulation. This functional decomposition improves readability and supports maintainability, and it also allows for easier adaptation of different steps.

Array based data structure: The core data structure used throughout the software are matrices and vectors stored as arrays. Arrays are stored in dense and sparse formats using NumPY and SciPY libraries, respectively (Harris et al., 2020; Virtanen et al., 2020). This choice reflects the mathematical and numerical formulation of the diffusion equation where discretised fields and time-stepping schemes are naturally represented by matrices. Mesh information imported from Gmsh (Geuzaine & Remacle, 2009) is also stored as arrays, maintaining the consistency. In some cases, Python dictionaries are used for convenience - e.g. grouping material properties - but do not affect the computational efficiency of the software, since the bulk of the computation is performed using Numpy arrays that support vectorization and multithreading. These data structures allow computationally efficient numerical operations and linear algebra computations, following a direct correspondence between mathematical formulation and implementation.

Software validation

Results of the software have been validated against published studies, measurements results, and external methods such as the acoustics radiosity method (Koutsouris et al., 2013) and the sound particle tracing method (Picaut & Fortin, 2012) and via informal auralization experiments. Integration testing of the main simulation engine (point 4 of the pipeline before mentioned) are in place to verify that the complete system functions seamlessly and that each of its submodules interacts correctly between each other. State verification against expected results are also implemented in the automatic tests. To complement the validation of the numerical results, the software is also profiled to ensure its robust performance under continuous changes.

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Research impact assessment

The software can be used as a standalone simulation tool or as a back-end acoustic simulator of the Community Hub for Open-source Room Acoustics Software CHORAS (Willemsen et al., 2025). The software has already been used in a number of scientific publications (Fichera et al., 2024, 2025, 2026) and graduate-level teaching at the Eindhoven University of Technology (TU/e), where it has proven valuable for understanding the impact of room geometry and material absorption on acoustic behavior and for designing rooms from an acoustics point of view. By combining ease of use and computational efficiency, `acousticDE` fills a need for researchers, practitioners, and educators seeking a practical, transparent and computational efficient tool for room acoustics simulation.

AI usage disclosure

No generative AI tools were used in the development of this software, the writing of this manuscript, or the preparation of supporting materials.

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